

BST Work-Package Ledger v1 — the single verified source of truth. One package per substrate object. When every package hits its OBJECTIVE, we are done.

Keeper | 2026-07-14 | Built from full parallel corpus verification reads (all 26 parameters + anchors, derived-vs-assigned pinned exactly). Authoritative tier source: the 26-Parameter Scorecard v0.3. THE MACHINE-READABLE LAYERS (**data/bst_constants.json**, **play/bst_26_table.py**) ARE STALE AND OVERSTATE CONFIDENCE — do NOT cite them; this ledger + scorecard v0.3 supersede. Rule: if the team cycles on any package, STOP and escalate to Keeper for re-analysis.

THE DEPENDENCY SPINE — everything funnels through 3 enablers + a few independents

Most open work is NOT 13 separate problems. It funnels through three enablers: - [K-TYPE QUANTIZATION] — the selection principle mapping each particle $\rightarrow$ its discrete K-type address (ν , and the two-label (k,m)) $\rightarrow$ localization norm N_i . The deep OPEN RESEARCH problem. Unlocks ~ 13 params: all CKM+PMNS mixing depths, the neutrino coefficients, the generation radii, and the muon/electron/tau address-forcing. Neutrino address $\nu=1/2$ is the ONE solved corner (F537). Charged/quark two-number selection is the frontier. Owner: Lyra. CASEY CLARIFICATION (2026-07-14) — resolve the tensor-vs-spinor object (CATCH 1) via the CONFORMAL ALGEBRA: work in the conformal algebra ($SO(5,2)$) and the (1,1)-tensor K-type MAPS to the spinors/fermions there — the conformal algebra IS the dictionary between the tensor slot and the $\Delta=2$ spinor/fermion content. So (1,1) and “the muon is a spinor” are NOT in conflict; the conformal algebra maps one to the other. Build that map $\rightarrow$ CATCH 1 resolves and the address-forcing is on solid ground. - [ENGINE C] — the boundary-integral evaluation capability (validated toy 4660; first task where the absolute normalization does NOT cancel $\rightarrow$ also the first real test of BST’s normalization conventions). Unlocks: muon residue ($3/8$ factor + Szegő=1), $\tau \sqrt{\pi}$ coefficient (forward), light-quark absolute anchor (m_u/m_d), α ’s 0.036 curvature term. Owner: Elie/Lyra. - [α KEYSTONE A] — the $U(1)$ gauge-coupling normalization: derive the count the kinetic term sees is exactly 137 AND the prefactor is unity. Unlocks: $\alpha \rightarrow$ DERIVED, and Higgs $\lambda=1/8$ (a passenger on the same law). Owner: Grace. - Independents: Majorana closure (γ^5 intertwiner + Cal co-sign); gen-1 up cold anomaly (Grace’s thermal lane); the cleanups (data-layer hygiene, the forks).

THE 26 WORK PACKAGES (by sector)

Format: TIER now $\rightarrow$ OBJECTIVE | derived | assigned/open | remaining work [enabler] | owner

Charged-fermion masses (9)

1. m_e — BY-DESIGN anchor $\rightarrow$ DONE (legitimate). The one dimensionful input (ℓ_B -Planck). $m_e=6\pi^5\alpha^{12}m_{Pl}$ is a 0.03% ratio-identification ($m_e/m_{Pl}\approx\alpha^{10.47}$, not clean — ratio-only theory). No work required; the anchor is legitimate.
2. $m_\mu/m_e = (24/\pi^2)^6$, 0.003% — PREDICTED (Casey F118 override “5”; objective strict count 4) $\rightarrow$ DERIVED. *Derived*: exponent $6=\dim SO(4)$, $64=d_\tau/d_\mu=2^{\wedge}\{C_2\}$, $R_{\{\Lambda^2\}}=Id$, $24/\pi^2=2^{\wedge}\{C_2\}/\text{vol}(S^4)$ exact, π -structure from the $\nu=3/2$ boundary Szegő residue (F117). *Assigned*: the flat $\rightarrow$ sphere $3/8$ factor; Szegő=1 (uncomputed); the address (1,1) (CATCH 1). CASEY CLARIFICATION (2026-07-14) — the $3/8$ IS physical, not a coordinate artifact: $3/(8\pi^2) = 1/\text{vol}(S^4) = 1/(2\cdot(4/3)\pi^2)$ — it is the factor from the muon passing through the boundary and “blowing up” into a particle. The $2\times$ = the two half- S^4 (boundary passage / $\mathbb{Z}_2$); the $(4/3)\pi^2$ connects to Casey’s 3-ball ($V(B^3)=4/3\cdot\pi$). So derive the $3/8$ as the boundary-transit / particle-formation normalization, NOT a flat $\rightarrow$ sphere conversion. Remaining: derive $3/8$ as boundary-passage + evaluate Szegő=unity [ENGINE C] + settle spinor-object [K-TYPE]. Owner: Elie+Lyra. Do NOT re-derive $(24/\pi^2)^6$ — it exists (F116/F117); close the two gaps.
3. $m_\tau/m_e = 49.71-\sqrt{\pi}$, 0.05% — NEAR-FORCED ($\sqrt{\pi}$ coefficient REVERSE-ENGINEERED) $\rightarrow$ NEAR-FORCED $\rightarrow$ PREDICTED. *Derived*: seat $\nu=0$, $\sqrt{\pi}$ -presence (odd n_C , F157). *Assigned*: the exact $-\sqrt{\pi}$ coef-

ficient (added to fix 7.7σ , retro-justified); 49-71 is an identification not mechanism; two competing forms (additive vs ratio). Remaining: evaluate the cone-tip residue forward, reconcile the forms [ENGINE C]. Owner: Elie.

4. $m_u = m_u/m_d = 1/\sqrt{g}$ (1.77 MeV) — IDENTIFIED → PREDICTED (capped ~20% by exp error). *Derived*: the ratio $1/\sqrt{g}$ exact at $n=0$; the inversion $m_u < m_d$ FORCED (proton stability). *Assigned*: two competing addresses (1.77 bulk vs 2.16 production); the gen-1 “cold anomaly” ($f(\nu)$ rank-2 → α^2 is $4.3\times$ off) HELD OPEN. Remaining: the $n=0$ deep-bulk overlap [ENGINE C] + the gen-1 thermal law (Grace, staged/unexecuted). Owner: Grace/Elie.
5. $m_d = n=1$ overlap (4.46 MeV, ~5%) — IDENTIFIED → PREDICTED (few-%). *Derived*: base rung of the F506 forced ladder; the $n=1$ value is a forward prediction (NOT a hand-set anchor, F511). *Assigned*: soft-est absolute; the “bare ribbon=current mass” premise is PI-endorsed, NOT Cal-ratified (if it fails, F506 reverts). Remaining: the $n=0,1$ overlap [ENGINE C] + Cal ratify the commitment/boundary premise. Owner: Elie/Cal.
6. $m_s = s/d = (N_c+1)(N_c+2) = 20$, 0.5% — BANKED (ratio) → DONE (ratio); absolute → PREDICTED. *Derived*: $\nu = N_c$ forced, {1,3,5} blind, single-row forced by Q^5 topology (toy 4618, closed the K671 gate). *Assigned*: the direction $\text{mass}(\nu)_\lambda$ not derived; absolute rides m_d . Remaining: derive the direction; absolute rides the anchor. Owner: —(banked).
7. $m_c = \alpha \cdot \nu / \sqrt{2}$, 0.05% — PREDICTED → BANKED when α derived. *Derived*: forward from $(m_p, m_e, g, N_{\max})$, zero charm input. *Assigned*: imports $\alpha = 1/N_{\max}$ (identified). Remaining: [α KEYSTONE A] closes it. Owner: —(rides α).
8. $m_t = (1-\alpha)\nu / \sqrt{2}$, 0.03% — PREDICTED → BANKED (leading) / DERIVED. *Derived*: ν forced with NO top input (F509, non-circular); $y_t=1$ derived two ways (C-S + point-fiber capacity). *Assigned*: the $(1-\alpha)$ correction is identified (rides α). Remaining: the leading value is essentially banked; the correction rides [α KEYSTONE A]. Owner: —(mostly done).
9. $m_b = b/d = 840$, but observed ~890 (5.6% OFF) — IDENTIFIED → IDENTIFIED until the 6% resolves. *Derived*: the ratio structure (same forced ladder as m_s , degree-5 rung). *Assigned*: the NUMBER misses 5.6% (scale/RG ambiguity — b and d quoted at different scales). Remaining: RG-run b/d between scales; $m_t/m_b = 42$ is an independent cross-check. Owner: Grace/Elie.

Neutrino sector (3 + nature)

10. $m_{\nu 1} = 0$ exactly — BANKED (=0) → DONE. Over-determined (chargeless odd-one-out + Z_3 + Möbius). Only mechanism-consolidation remains.
11. $m_{\nu 2} = (7/12)M_0$, 0.35% — IDENTIFIED → DERIVED. *Derived*: the seesaw SHAPE $m_{\nu} \propto \alpha^2 m_e^2 / m_p$ (dimensionless, given the ansatz). *Assigned*: coefficient 7/12 FITTED (three incompatible forms; formal-degree route RULED OUT, toy 4658); α^2 power justification (Hopf-vertices) ORPHANED by the Dirac → Majorana flip. Remaining: force 7/12 via the Bergman Majorana-matrix diagonalization [K-TYPE/ENGINE B]; only lead = fiber-overlap ~4%. Owner: Elie.
12. $m_{\nu 3} = (10/3)M_0$; Δm^2 ratio $(40/7)^2 = 1600/49$, 0.3% — IDENTIFIED → DERIVED. *Assigned*: coefficient 10/3 FITTED; Δm^2 ratio has a 3-way theorem fork (32.65/33/34) — adjudicate (cheap). Remaining: force 10/3 [K-TYPE/ENGINE B] + resolve the fork. Owner: Elie + Keeper(fork).
13. Nature = MAJORANA — NEAR-FORCED / forward-from-address (Dirac RETIRED) → BANKED. *Derived*: $\nu = 1/2$ forced → non-unitary shadow $\nu = 9/2$ → no ν_R (F537); BOLT-1 rigorous; F331 pseudoreal symplectic-Majorana solid. *Assigned*: two joints — the explicit γ^5 intertwiner construction (toy 4659 built the reflection; rigor pending) + Cal’s co-sign (required to flip a banked falsifier). Remaining: both joints → flips pred_004 to a 1–4 meV $0\nu\beta\beta$ floor. Owner: Elie+Cal. Independent of masses/mixings — near-term closable.

CKM mixing (4)

14. $\theta_{12} / V_{us} = 1/\sqrt{20}$, 0.4% — IDENTIFIED/LEAD → DERIVED. *Derived*: the Fritzsch texture-zero location ($d(5/2)=0$) + banked $s/d=20$. *Assigned*: the kernel does NOT saturate ($K_{12}=0.237 \neq 1$); $1/\sqrt{20}$ re-expresses $N_2 = 20^{-1/5}$, doesn’t force it. Remaining: derive $M_{12} = \sqrt{(m_d \cdot m_s)}$ forward, OR force the gen-2 address [K-TYPE]. Owner: Grace/Lyra.

15. $\theta_{23} / V_{cb} = 0.041$, angle $\cos\psi = 5/\sqrt{34}$ — NEAR-FORCED (angle DERIVED, magnitude $\sim 7.5\%$ structural) $\rightarrow$ DERIVED. *Derived*: the ANGLE (target-innocent, primaries-only; $\mu @ \hat{e}_1$ reduces to “deposit=dilation axis”). *Assigned*: the MAGNITUDE ($\sim 7.5\%$) gated on r_τ ’s exact K-type address + the A/B seam (unreconciled) + $f(\nu)$ up-weight (the blocker). Remaining: pin r_τ [K-TYPE] + $f(\nu)$ + reconcile A/B. Owner: Grace/Lyra.
16. θ_{13} / V_{ub} — OPEN (build): $V_{ub} = V_{us} \cdot V_{cb} \cdot \text{apex}$, $\text{apex} = 0.414$; pairwise $21\times$ off $\rightarrow$ OPEN. *Derived*: only the form (not-independent, hierarchy-tied). *Assigned*: magnitude doubly-gated (CP-complex-positions + gen-1 up anomaly). Remaining: the full $U_u^\dagger U_d$ complex build [K-TYPE/ENGINE B]. Owner: Grace+Lyra.
17. δ_{CKM} — OPEN (can-fail): $J=0$ for real localizations PROVED (F498) $\rightarrow$ OPEN. *Derived*: only “CP exists because n_C odd” (structural, IF peaks are complex). *Assigned*: ALL concrete δ values are reverse-fits ($\arctan\sqrt{5}$, 309° , $3\pi/7$, π). F533’s forward $\delta = \arctan\sqrt{5}$ is DEMOTED (see CP adjudication below). Remaining: the explicit complex generation-state build $\rightarrow$ compute J; can genuinely FAIL [K-TYPE/ENGINE B]. Owner: Grace+Lyra.

PMNS mixing (4)

18. $\theta_{12} = |U_{e2}|^2 = 3/10$, 0.06% — IDENTIFIED (demoted from BANKED; 42/137 was a coincidence) $\rightarrow$ DERIVED. *Derived*: the matrix-element EXTRACTION ($\div \cos^2\theta_{13}$) forced; 3/10 un-cheapened vs 5/16. *Assigned*: “why $|U_{e2}|^2 = 3/10$ ” (the overlap value) OPEN. Remaining: forward the dual- ρ 1-2 overlap [K-TYPE/ENGINE B]. Owner: Grace/Lyra.
19. $\theta_{13} = 1/45$, 1.2% — BANKED (only PMNS bank) $\rightarrow$ DONE (bank); forcing-theorem open. *Derived*: target-innocent, π -free, convention-robust, $|U_{e3}|^2$ stable. *Assigned*: the “why $N_c^2 \cdot n_C$ dimension-count” forcing is NOT a closed theorem (K229b). Remaining (to upgrade the forcing): close the dimension-count [K-TYPE]. Owner: —(banked).
20. θ_{23} — IDENTIFIED, octant OPEN: 4/7 vs 6/11 (both upper-octant) $\rightarrow$ NEAR-FORCED (DUNE-gated). *Derived*: upper octant predicted (falsifier). *Assigned*: form unstable (4/7 vs 6/11), neither forced; the $\times \cos^2\theta_{13}$ “fix” was withdrawn as fished. Remaining: force the form [K-TYPE] + DUNE resolves the octant. Owner: Lyra + experiment.
21. δ_{PMNS} — OPEN (all prior values retired reverse-fits) $\rightarrow$ OPEN $\rightarrow$ PREDICTED. *Derived*: only the falsifier $\delta \neq \{0, \pi\}$ FORCED (odd n_C). *Assigned*: no forward value; $\sim 197^\circ$ hint consistent. Remaining: derive the dual- ρ complex phase [K-TYPE/ENGINE B]; DUNE scores. Owner: Lyra + experiment.

Gauge couplings (3)

22. α — $\square$ COMPLETE (Casey ruling 2026-07-14: “mark alpha complete, the $[4\pi]$ convention question is trivial at this point”). *Derived + banked*: the count $137 = 27 \cdot n_C + \text{rank}$; the coupling = $1/\text{count}$ (democracy — S^1 Haar equal-norm, COMPUTED); the scale-falsifier PASSED (no stray $2\pi/c_{FK}/4\pi$); the 4π convention judged a trivial natural normalization (Casey). The finite-capacity “why” (compact boundary $\rightarrow$ capacity $\rightarrow$ reciprocal) grounds it. $\alpha = 1/137$ is DERIVED. *Separate precision refinement (does NOT block completion)*: the 0.036 second-order term = an Engine-C heat-kernel integral (leading $n_C/N_{\max}$ 1.4%); this is a precision add-on to $\alpha^{-1} = 137.036$, not part of the “why $\alpha = 1/137$ ” derivation. Owner: — (complete; 0.036 optional precision via Engine C).
23. $\sin^2\theta_W = 3/13$ — BY-DESIGN runner $\rightarrow$ DONE (legitimate by-design). A clean value-form at one scale; running is standard RG. Not a failure. Objective: honest by-design.
24. $\alpha_s = 7/20 @ m_p$ — BY-DESIGN runner $\rightarrow$ DONE (legitimate by-design). Value + $c_1 = 3/5$ (3 proofs). Scale-pin + non-pert running by-design.

Higgs (2) + strong-CP (1)

25. $v = m_p^2/(g \cdot m_e)$, 0.04% — BANKED (ratio) / FLOOR (absolute) $\rightarrow$ DONE (ratio). *Derived*: full $m_{Pl} \rightarrow m_e \rightarrow m_p \rightarrow v$ chain, NO top input (F509, verified non-circular). *Assigned*: $g=7$ mechanism form-pinned; absolute scale = the one dimensionful input. Remaining: derive the $g=7$ mechanism; absolute stays FLOOR. Owner: —(ratio banked).

26. $m_H / \lambda = 1/2^{N_c} = 1/8 \rightarrow v/2, 1.7\%$ — IDENTIFIED/LEAD $\rightarrow$ DERIVED when α Keystone A closes. *Derived:* v forced; the $1/8$ dilution mechanism (α -parallel, target-innocent). *Assigned:* passenger on $[\alpha$ KEYSTONE A] (show the spinor kinetic term sees exactly 8, prefactor unity); the 1.7% correction held-not-fitted; a competing form $\sqrt{(2/5!)}=0.11\%$ exists (form not uniquely forced). Remaining: Keystone A (λ -side) + adjudicate $1/8$ vs $\sqrt{(2/5!)}$. Owner: Grace/Lyra.
27. $\theta_{QCD} = 0$ exactly — BANKED (founding) $\rightarrow$ DONE. *Derived:* T1964 contractibility $\rightarrow$ trivial bundle $\rightarrow \int c_2=0$. *Assigned:* “ $\pi_1=0$ ” is imprecise shorthand — the load-bearing statement is CONTRACTIBILITY $\rightarrow$ trivial bundle; make it explicit (rigor-formality, not a value question). Owner: —(banked).

(26 SM params + m_e anchor = the full set; $m_{\nu 1}$ /nature and v -ratio/ v -abs split some rows.)

COMPLETION STATUS (objective, honest)

- DONE — banked/forced or legitimate by-design (11): α (COMPLETE — Casey ruling; the 0.036 is optional precision, not blocking), m_e (anchor), m_s (ratio), $m_{\nu 1}=0$, θ_{13} PMNS, v (ratio), θ_{QCD} , $\sin^2\theta_W$ (by-design), α_s (by-design), m_t (leading). Note: m_c and m_t ’s $(1-\alpha)$ correction now inherit a COMPLETE $\alpha \rightarrow$ both firm up.
- NEAR-DONE — predicted/near-forced, gated on an enabler (6): m_μ/m_e ($3/8+\text{Szegő}+\text{address}$), m_τ/m_e ($\sqrt{\pi}$ coeff), m_c (rides α), Majorana nature (2 joints), θ_{12} PMNS (overlap value), V_{cb} (magnitude).
- IDENTIFIED — real work, gated on an enabler (7): m_u , m_d , m_b , $m_{\nu 2}$, $m_{\nu 3}$, V_{us} , m_H/λ .
- OPEN — build required, can-fail (4): V_{ub} , δ_{CKM} , θ_{23} octant (DUNE), δ_{PMNS} .

When are we DONE? When these close: [K-TYPE QUANTIZATION] (unlocks the mixing sector + ν -coefficients + generation depths ≈ 13), [ENGINE C] (muon $3/8$ +residue, $\tau \sqrt{\pi}$, light-quark anchor, α 0.036), [α KEYSTONE A] ($\alpha + \lambda$), the Majorana joints, the gen-1 up anomaly, and the honest by-design items are accepted as endpoints. That is a FINITE, named set — not an open horizon.

CRITICAL — DATA-LAYER / SCOREBOARD ARE STALE (extends K684; hygiene gate before ANY outreach)

The machine-readable layers CONTRADICT the honest scorecard and OVERSTATE confidence: - **data/bst_constants.json**: α at tier-D “Wyler integral”; CKM CP at tier-D “ $\arctan\sqrt{5}$ ”; quark rows const_106–110 carry legacy tier-D fitted forms (NOT the current $s/d=20$, $1/\sqrt{g}$, $\alpha \cdot v/\sqrt{2}$); PMNS $\theta_{12}=4/13$ tier-D; PMNS CP= $3\pi/7$ tier-D. A .bak_2026-07-14_grace_K684_demotions exists (demotions in flight) but the live rows still read stale. - **play/bst_26_table.py**: α still the Wyler closed form (line 37); θ_{QCD} “ $\pi_1=0$ ” shorthand (line 84 self-flags). - ACTION [Grace owns data/]: sync bst_constants.json + bst_26_table.py to scorecard v0.3 BEFORE any verify_bst.py-based outreach. The scorecard v0.3 + this ledger are the ONLY authoritative tier sources.

CP ADJUDICATION (resolves a same-day conflict)

Lyra F533 asserts a FORWARD $\delta_{CKM} = \arctan(\sqrt{5}) = 65.9^\circ$ (odd- n_C). K683/K684 treat $\arctan\sqrt{5}$ as a reverse-fit and δ as OPEN. Verdict (Keeper): the QUALITATIVE claim “CP violation exists because n_C is odd” is target-innocent and survives (structural, conditional on the peaks being complex — F498). The QUANTITATIVE $\arctan\sqrt{5}$ is exactly the reverse-fit K684 flagged (three incompatible “derived” δ values is the proof). δ_{CKM} stays OPEN (can-fail); the value is NOT forward. F533’s forward- δ claim is DEMOTED to “CP-exists structural + δ -value open.” Same for δ_{PMNS} (only the $\neq\{0,\pi\}$ falsifier is forward).

CYCLE-PREVENTION RULES (per Casey)

1. This ledger + scorecard v0.3 are the source of truth. Before working any object, read its package here — do NOT re-derive what's marked derived (the muon $(24/\pi^2)^6$ is the cautionary case: it exists in F116/F117; close the named gaps, don't reopen).
2. Each package states its OBJECTIVE. Work toward that classification; when hit, mark it and stop.
3. If you cycle (re-deriving, or a form/value oscillates $\geq 2\times$), STOP and escalate to Keeper for re-analysis (per Casey's instruction). The forks (θ_{12} , Δm^2 , $m_{H-\lambda}$, τ -form) are cycle-risks — Keeper adjudicates, the team doesn't re-litigate.
4. Do not bank across a gate: the “banked 5” muon is a Casey override, not an objective forward derivation — objective count is 4 until 3/8 + Szegő=1 + the spinor-object land.

— Keeper, 2026-07-14. Work-Package Ledger v1. 27 packages (26 SM params + m_e anchor). 10 done, 6 near-done, 7 identified, 4 open. Everything funnels through K-TYPE QUANTIZATION + ENGINE C + α KEYSTONE A + Majorana joints + the gen-1 anomaly. Data-layer/scoreboard STALE — scorecard v0.3 + this ledger are authoritative. See [[Keeper_26_Parameter_Scorecard_v0.1_2026-07-12]], [[Keeper_K682...]], [[Keeper_K685...]], [[Keeper_K684...]].